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SRv6 design in large SPs networks

Architecture journey

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01 Iliad Group and Free Who are we?

The parent company of Free

Founded in 1990, the Iliad Group is a major player in the European telecoms sector.
It is the parent company of Free, the French trouble maker that has revolutionized telecoms.





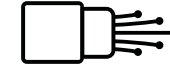
12,128
EMPLOYEES
France & Morocco



22 million
SUBSCRIBERS

14.8 million
Mobile
subscribers

7.3 million
Fixed
subscribers



34.3 million
MARKETABLE
FIBER SOCKETS



23,524
MOBILE SITES



220
STORES

The Group in Europe



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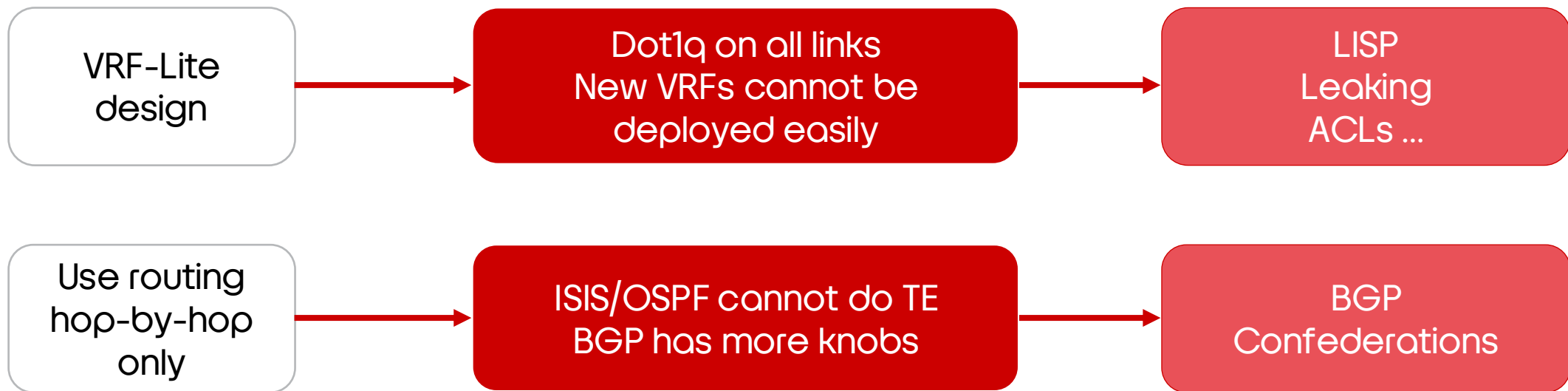
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02 The origins

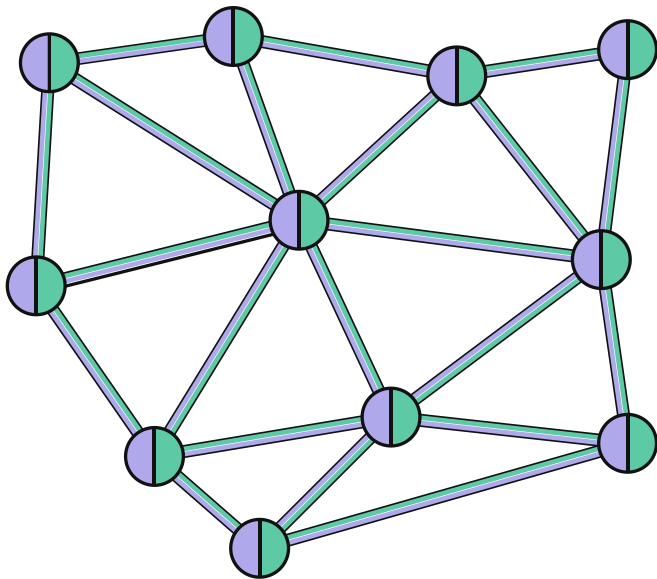
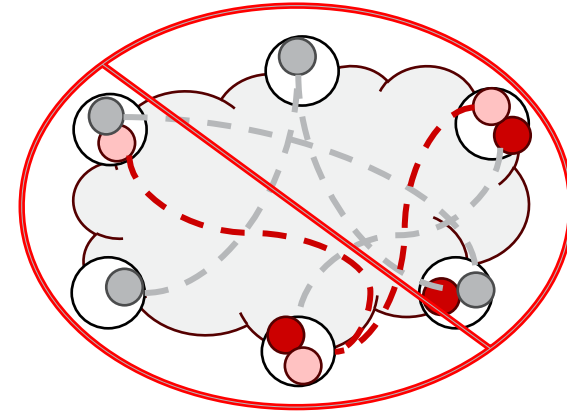
Historical Backbone (France)

- Purely IP based
- Service & Transport needs evolved, and required solutions

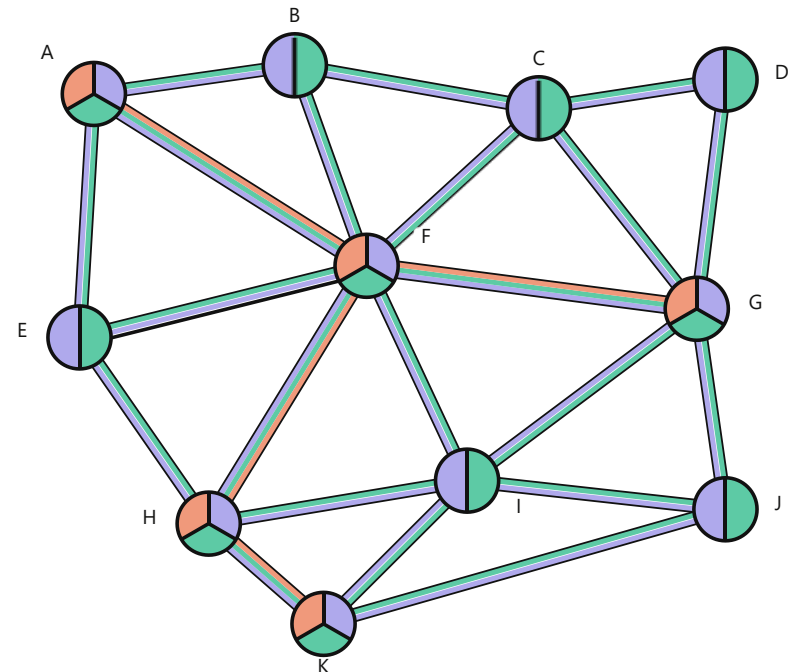


Historical Backbone: VRF mgmt

No flexible VRF placement

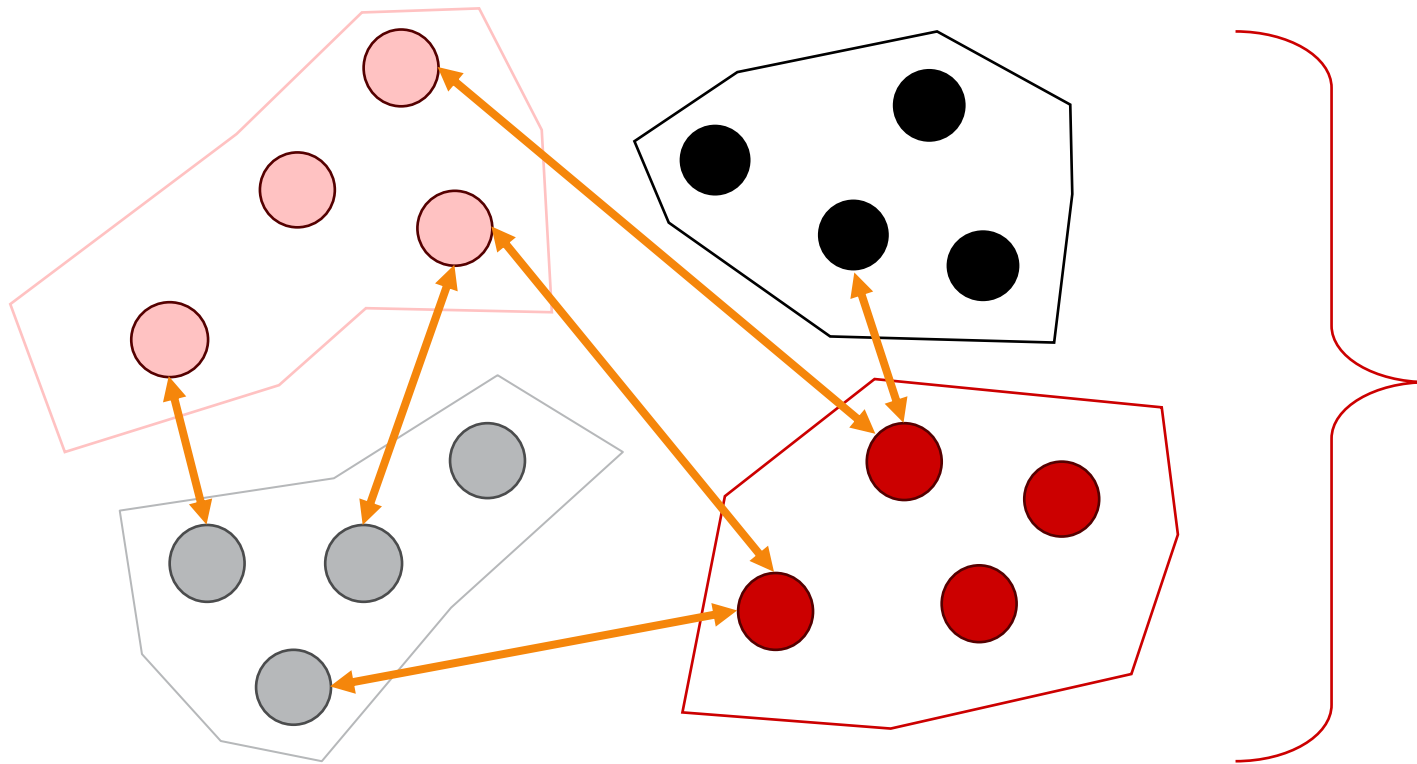
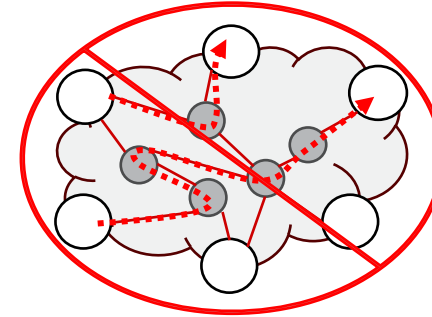


very cumbersome!



Historical Backbone: traffic steering

No tunneling,
no advanced
TE



Confederation architecture allows **traffic steering** via BGP attribute manipulation at red links

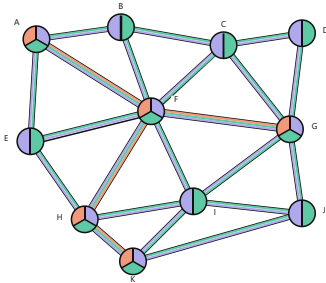
When SRv6 appeared, its deployment followed the same principles ...

SRv6 introduction

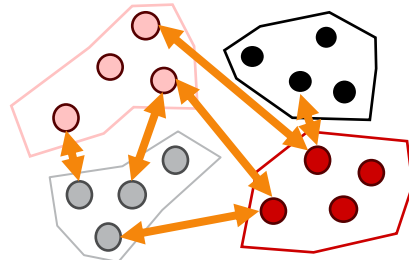
- MPLS was difficult to deploy in parallel of confederations
- SRv6, being IP-based, was much easier
- Iliad Italia's backbone was 100% based on SRv6

Transport

Default VRF =
additional VRF-lite

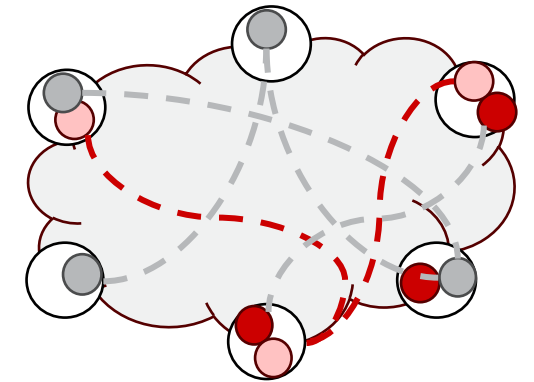


Confederations kept
for traffic steering ⇒
no global IGP for
default



Service

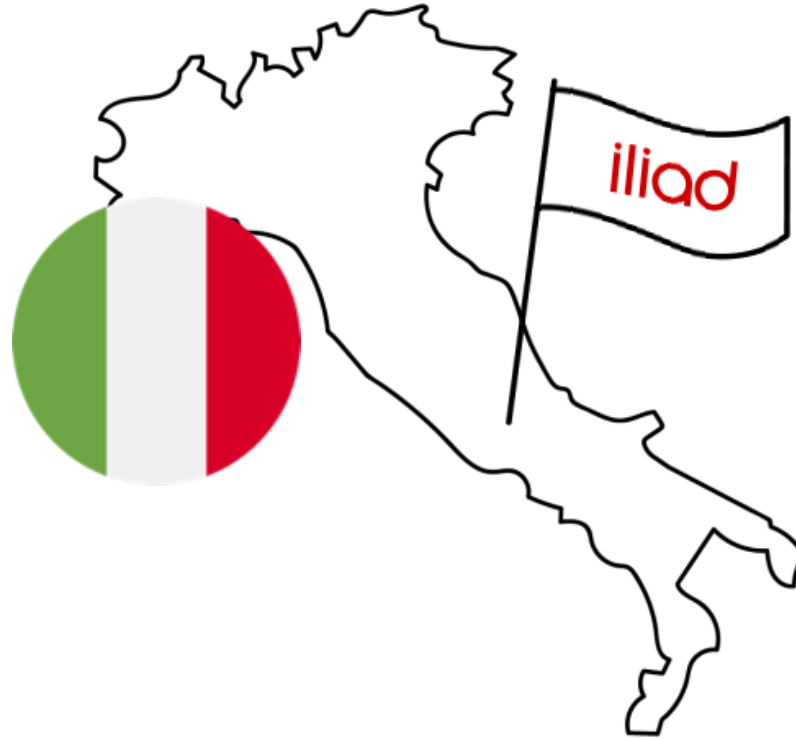
Service is end-
to-end (no
NHS)



Like « inter-AS option C »
Though not sticking to textbook

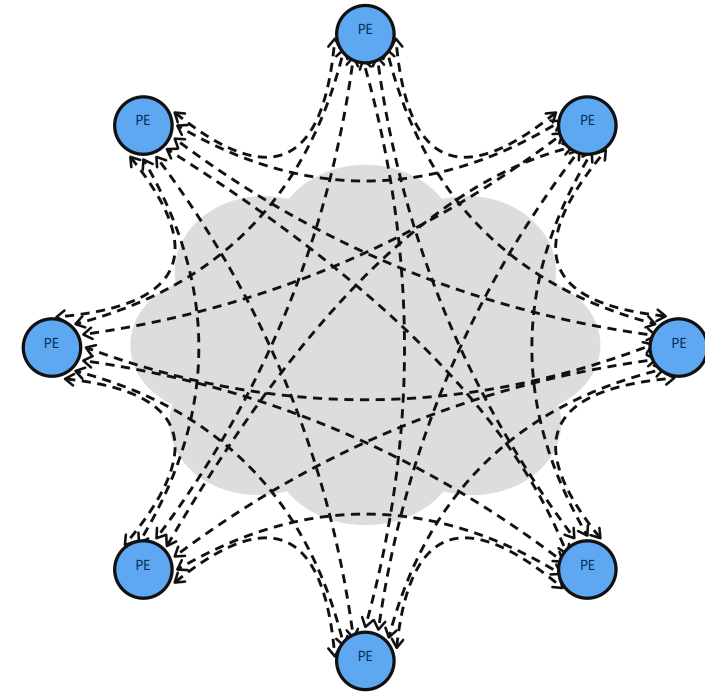
Iliad Italia – SRv6 Full-Length SID

- 12,000 cell-site-routers
- 1,700 backbone routers
- Default VRF: BGP confederations



Towards the New Backbone (France)

- Key tech points:
 - > Have an overlay/underlay VPN design based on SRv6 uSID
 - > Have a single IGP domain
 - > Get rid of confederations, use light TE or just metrics
- General design principles:
 - > End-to-end service delivery (**service-stitching**)
 - > **Sub-50ms fast-reroute**
 - > Simple protocol stack (**BGP-free core**)
- Benefits expected:
 - > Operational simplicity
 - > Flexibility for growth and transformation



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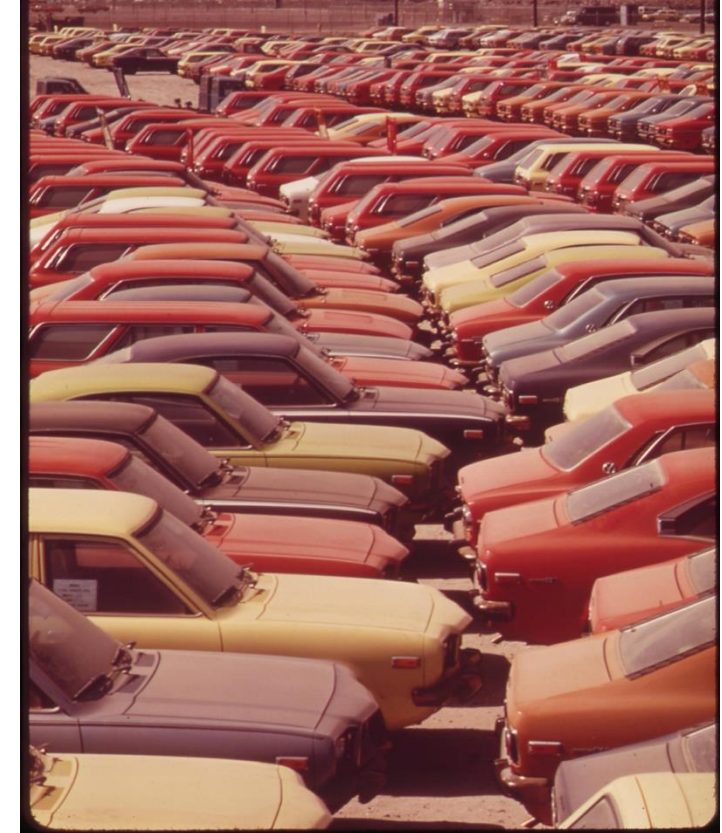
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03 The scale is the limit
Or is it?

The scale discovery

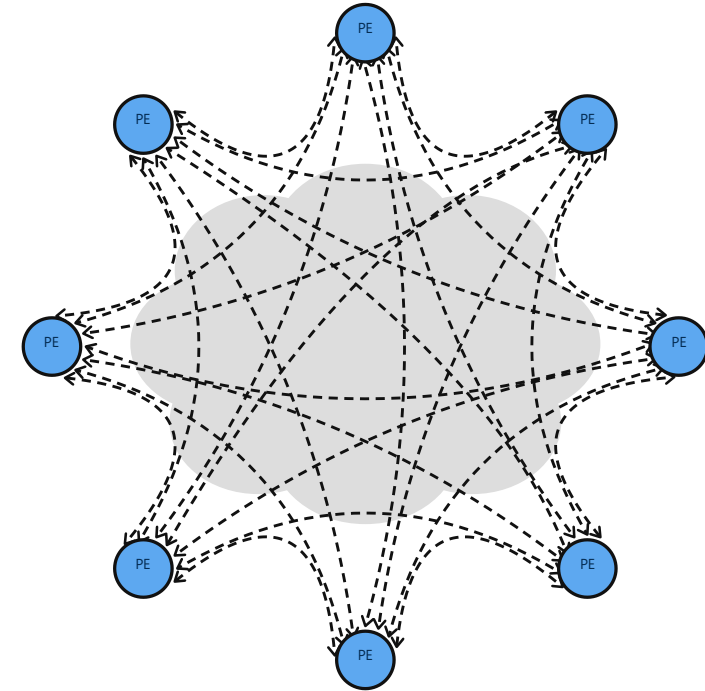
- One day, one router reached ***Out Of Resources*** (OOR) on **SRv6 encap scale**
- **Encapsulation scale was then identified** as paramount in our designs
- Situation is more important with SRv6 (vs SR-MPLS) because open source software widely supports SRv6 now - remember our **homemade Cell-Site-Routers**
- **Cisco** worked to **help** us:
 - > Custom development of a **new hw-profile (NCS/J2) for SRv6 service scale increase**
 - > Cisco 8000 routers NPU scale increase : K100 tested against numbers well-above industry average



Well ... no spot left.

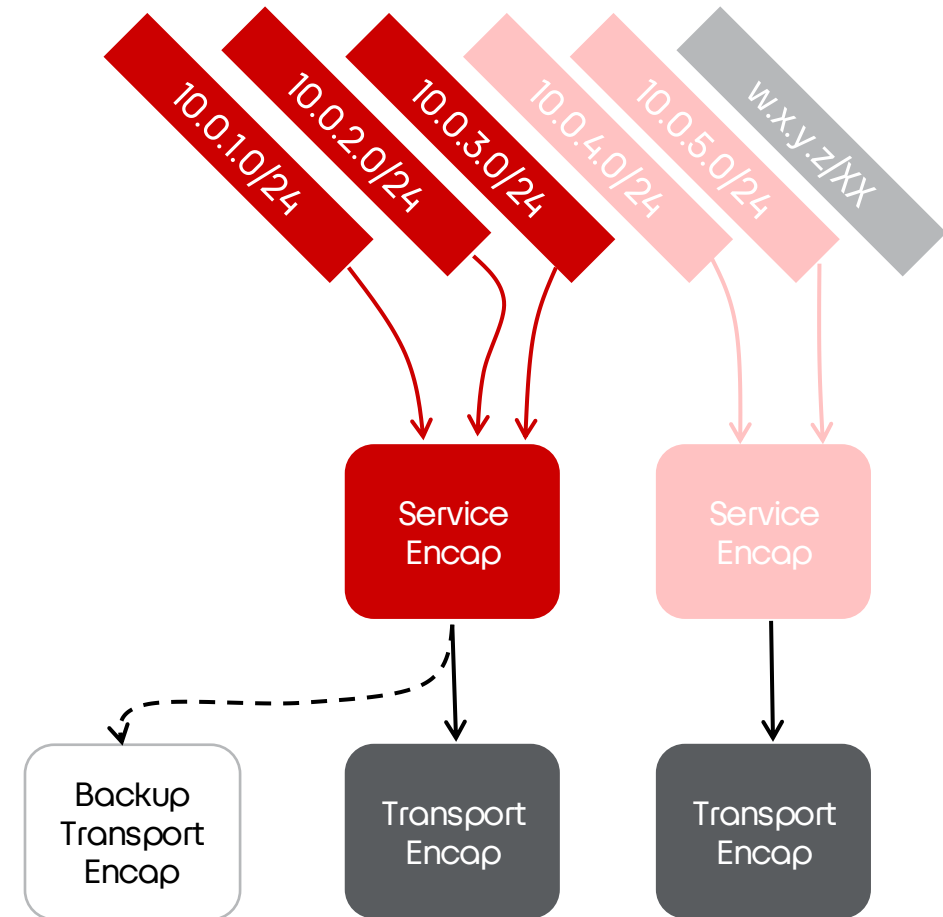
Scale on NBB (2030+)

- **Backbone:** 3,000 nodes
- **Cell Site Router:** 25,000 nodes
- **Optical Line Terminal:** 12,000 nodes
- Today:
 - > OLT: layer2 (no SRv6)
 - > CSR: SRv6 in progress (separate ASN, BGP-locators, IAS option C)
 - > Transport: confederation
- Tomorrow:
 - > OLT: layer2 at first but SRv6 possible
 - > CSR: all SRv6 uSID
 - > Transport: SR-ISIS single or multi-domain



VPN next-hop

- In theory, forwarding a packet from CE to VPN core looks like:
 1. IP lookup in VRF table
 2. **VPN-NH is found** and needs an **additional encapsulation before packet is sent: MPLS or SRv6**
 3. No other lookup required – packet is sent out in default VRF
 4. (other stages such as load-balancing, L2 encap, etc, can happen)
- **Hierarchical FIBs** allow hardware Fast-Reroute aka « Prefix Independent Convergence »:
 - > Use separate objects and update pointers
 - > Typically used for Encap where Service and Transport can be decoupled



Scale optimization – what to look at

– Transport:

- > Fast-reroute (TI-LFA/LFA) tunnels
- > More loopbacks/prefixes in IGP consume more
- > Large IGP: trigger big HW re-programming when failures occur
- > Traffic-Engineering consumes for each policy – but we will use only a handful

– Service:

- > More remote VRFs = more remote SIDs to store
- > HVPN usually based on service prefix aggregation

SRv6 vs MPLS

SRv6 transport is backwards-compatible with IP

Longest-Prefix-Match forwarding

Transport aggregation is possible
⇒ Locator Summarization

SRv6 service enhanced

Cisco's collaboration
SRv6 Gateway qualified (available)
SRv6 HVPN upcoming

Per-L3VPN route HVPN treatment+AI GP
stamping
⇒ Massive scale reduction

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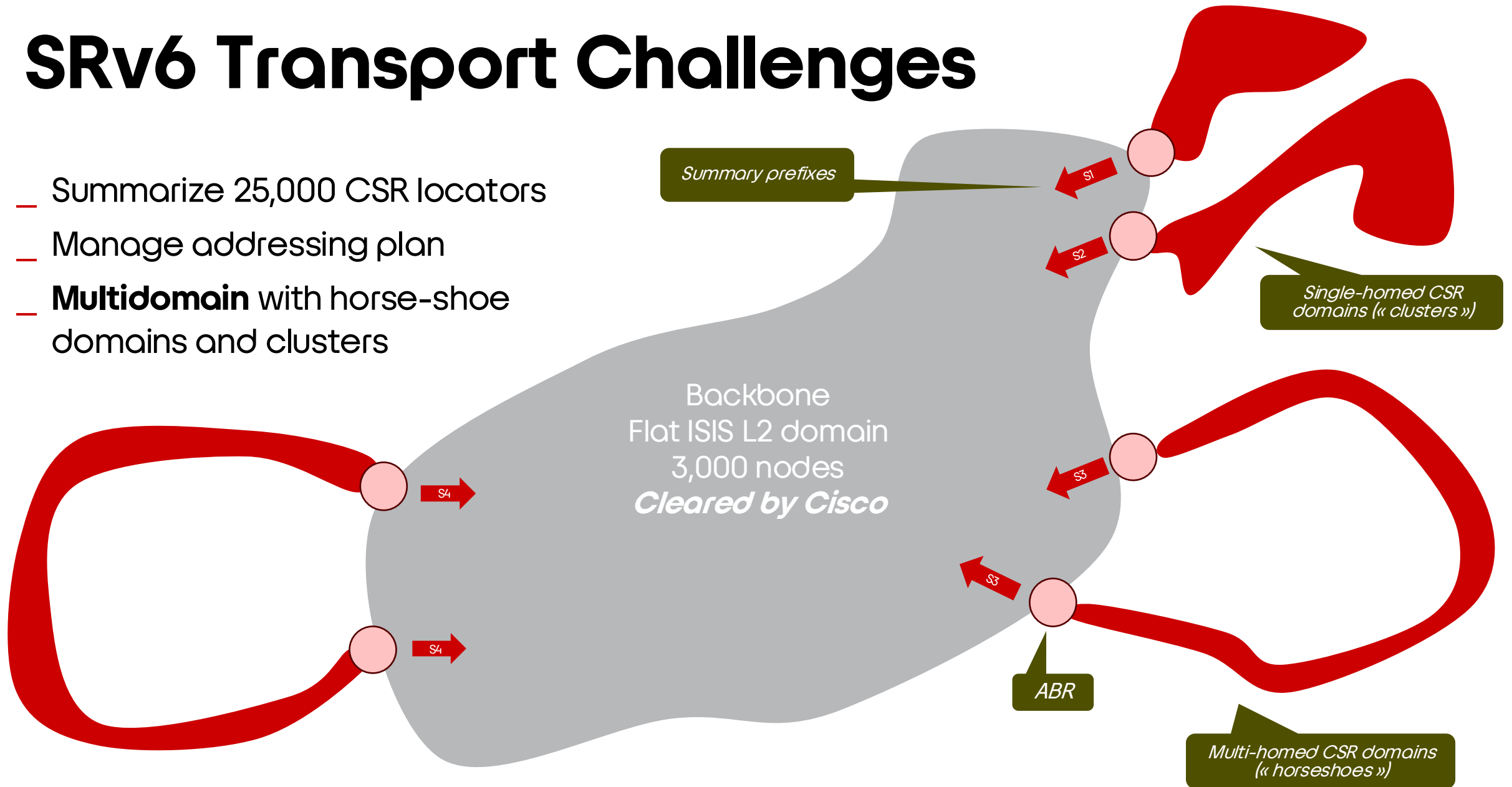
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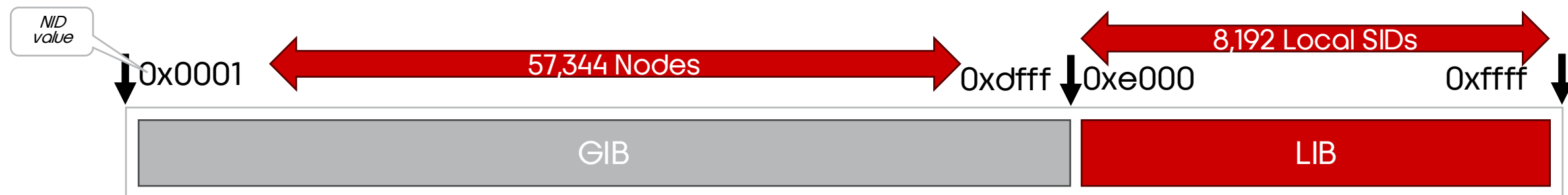
04 Designing our new SP backbone

SRv6 Transport Challenges

- Summarize 25,000 CSR locators
- Manage addressing plan
- **Multidomain** with horse-shoe domains and clusters



F3216 planning for summarization

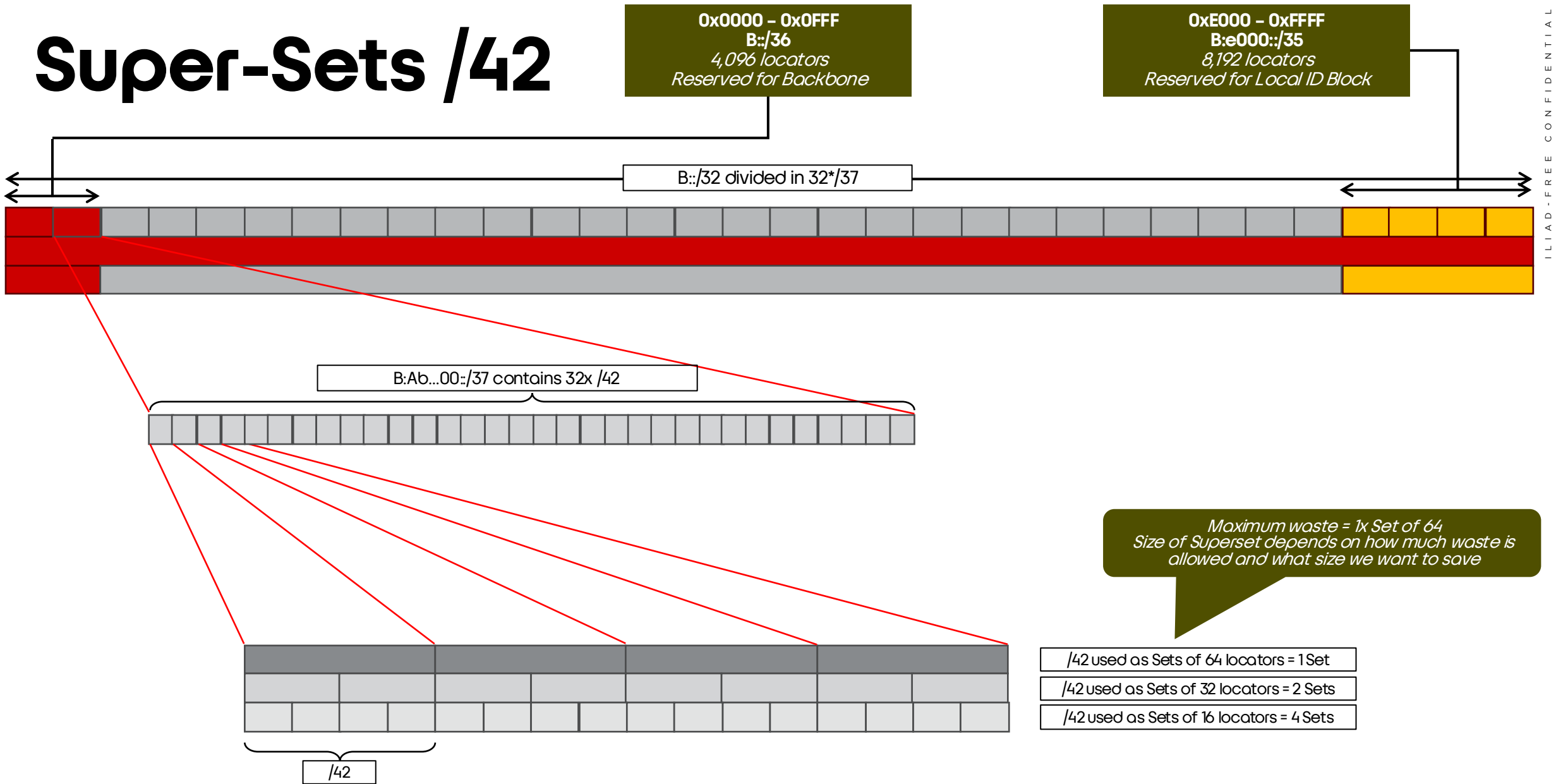


- Fit all in 1 base block /32
- Adapt to varying CSR island sizes
- Avoid address space fragmentation

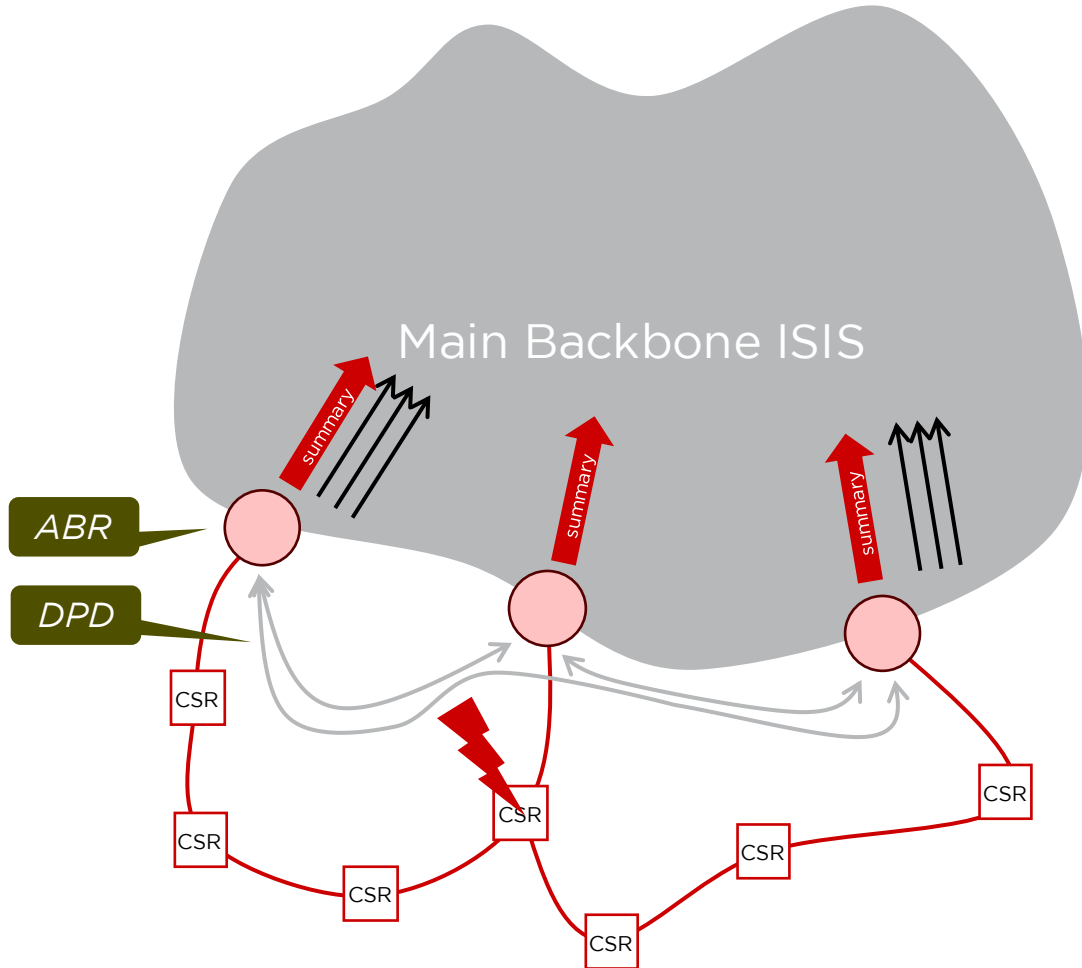
Use ***super-sets*** blocks that can only be divided into sets of the same size, avoids waste

Set size	Node count	Set count
/46	4	14,336
/45	8	7,168
/44	16	3,584
/43	32	1,792
/42	64	896

Super-Sets /42

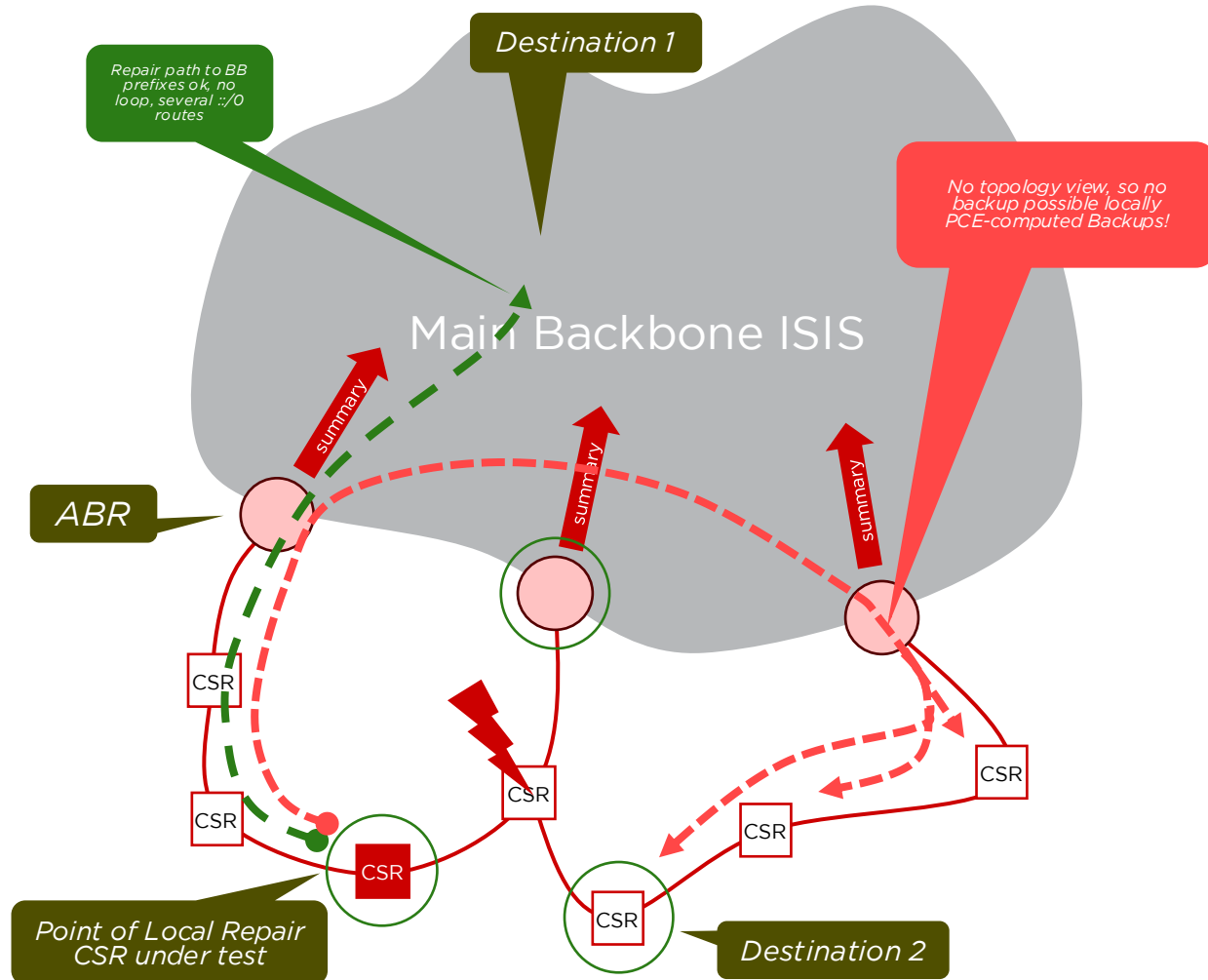


Multi-homed islands: DPD



- **Nominal**: each ABR sends summary-prefix
- **CSR fails**: Domain-Partition-Detection triggers, and all ABRs start redistributing individual locators
- *This covers for **global convergence**, eg. it restores communication after a few seconds*

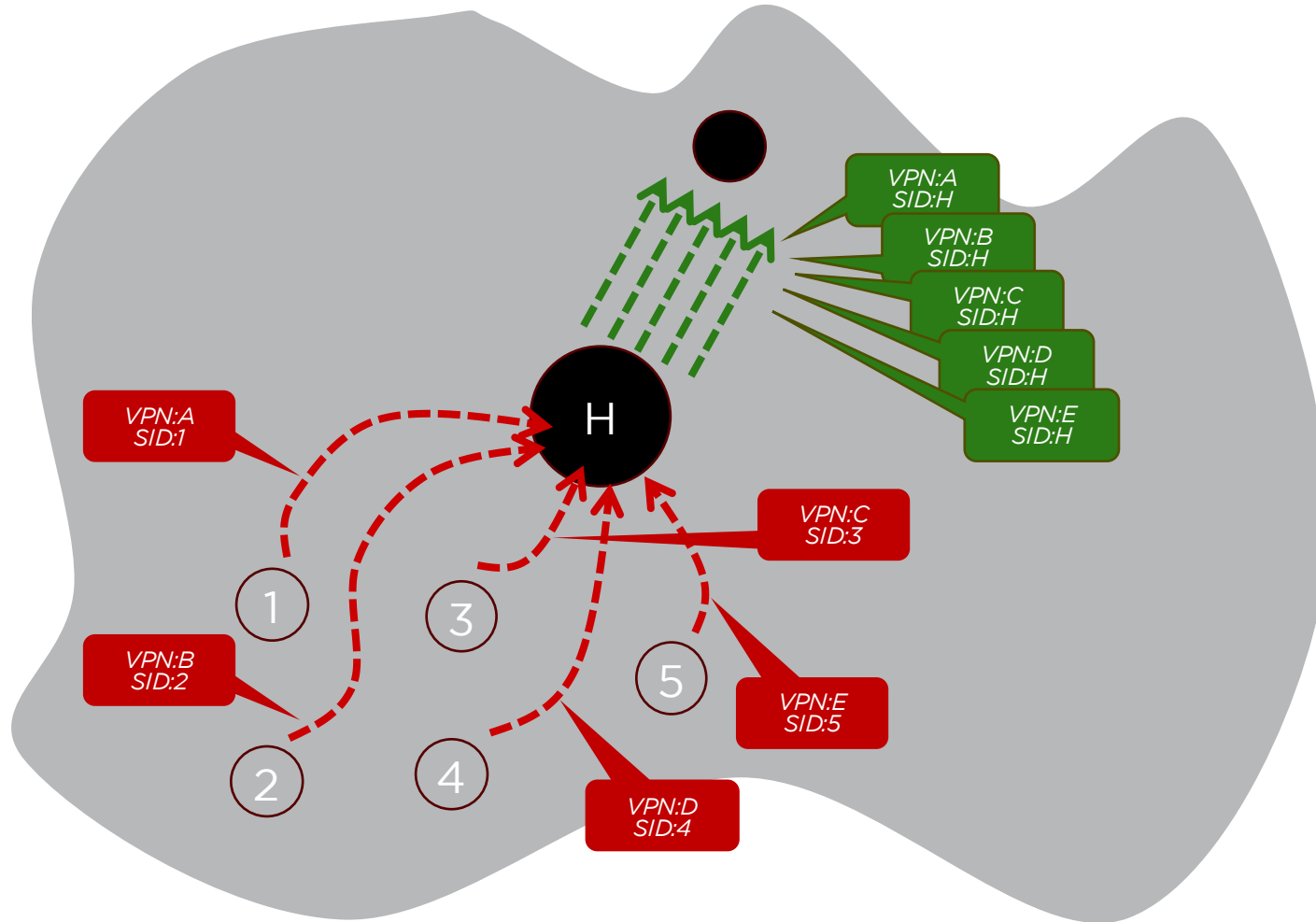
Multi-homed islands: FRR



From PLR (under test) standpoint

- **Local (TI)LFA** computation is possible for $::/0$
- But **topology is not redundant** for local nodes
- Without BB view, « loop free » cannot be guaranteed
- *Hence, using a PCE to query for (TI)LFA stack is a good solution when crossing several domains*

SRv6 Hierarchical VPN



- Usual HVPN requires Aggregates at VPN level
- Very hard in brownfield & operations
- HVPN feature for SRv6: re-originate same routes with a single SID
- Creates waypoint & tunnel termination – tradeoff between E2E and scale
- Many other usecases:
 - > Full SID/uSID migration
 - > MPLS/SRv6 migration
 - > FlexAlgo conversion

SRv6/SRv6 Gateway: available & tested for SRv6/SRv6 scenario by Cisco

HVPN: H1-2027

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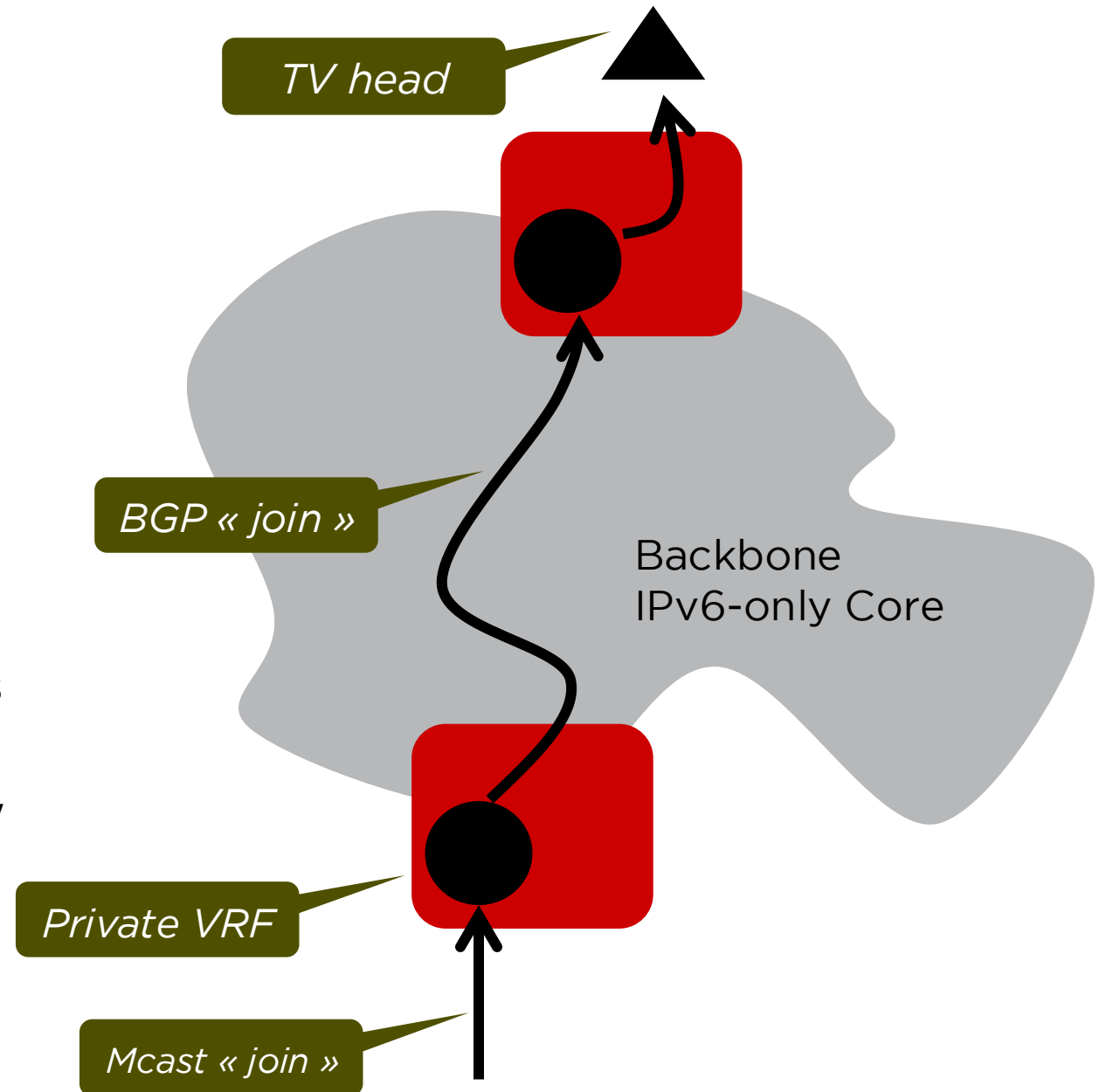
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05 Multicast TV distribution

Multicast for TV

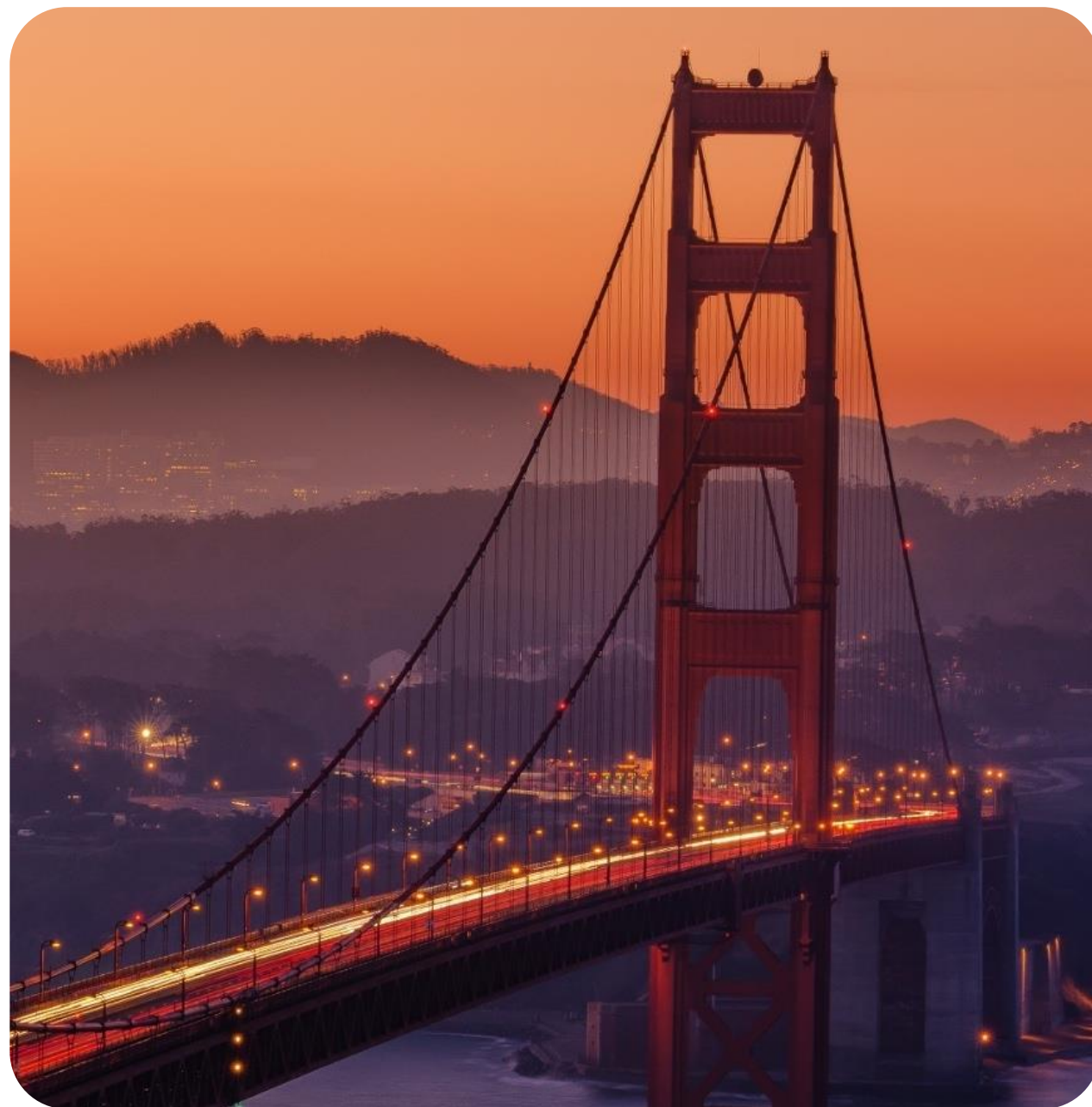
- Multicast inside a VRF? NG-MVPN
- Over IPv6 core? No mLDP, no Rosen-GRE IPv6 ...
- NG-MVPN for IPv6 cores by Cisco: IPIP PMSI encap!
- Bonuses:
 - > Use MoFRR for better convergence times (anycast PE)
 - > Shard $n*(C-S, C-G)$ into $1*(P-S, P-G)$ for bw efficiency



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Takeaways

- SRv6 scales better than MPLS, by design
- SRv6 toolbox is nearing maturity for SPs
- Many existing concepts can be strategically reused to make multi-domain operationally easy



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